

Evidence Sources, Trace Reconstruction, and Evaluation Metrics

Evidence Sources

Data Collection & Trace Reconstruction

1.1 Evidence sources and provenance

 GitHub Issue resolution through Agents

SWE-bench Pro
(584 run-level trajectories)

 **Claude 4.5 Sonnet**
(Anthropic)
319 trajectories

 **GPT-5**
(High)
265 trajectories

DeepSWE v1.1
(top-10 leaderboard configurations)

 **Claude Opus 5**
(high / max / xhigh)

 **GPT-5.6 Sol**
(max)
4 configurations × 327 = 1,308 run-level trajectories
(success-conditioned)

SWE Agent harness trajectories Logs

1.2 Trace reconstruction and information boundary

C Context acquisition
- issue & repository context
- retrieve snippets
- ...

S Search / exploration
- search queries
- tool usage
- candidate context (ranked)
- ...

G Grounding
- opened files
- evidence links
- candidate justification
- ...

V Verification
- test/build results
- error messages
- fix validation
- ...

A Audit / controller
- audit decisions
- next action selection
- tokens / calls
- ...

Canonical Agent Trace τ
 $\tau = (C, S, G, V, A)$

1.3 Outcomes, cost, and statistical procedures. Relating Process Quality to Outcome Necessity

Canonical Agent Trace τ

Grounding
(q_{ground})


Path Quality
(q_{path})

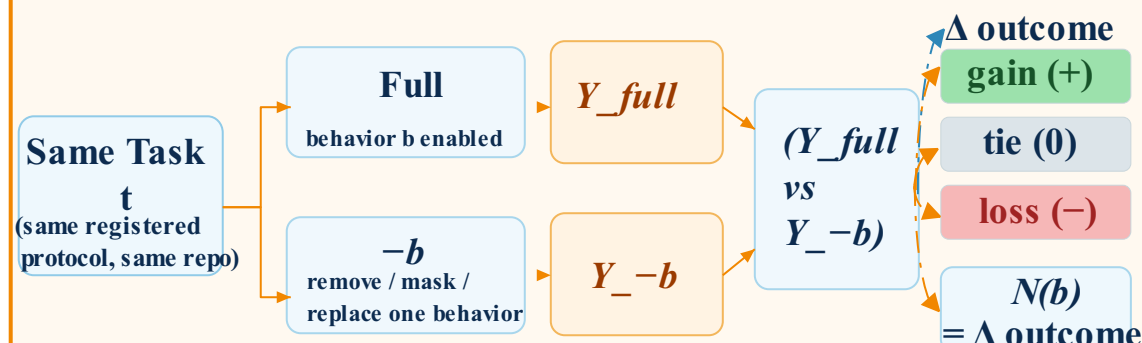

Verification
(q_{verify})


Coherence
(q_{coh})


Efficiency
(q_{eff})


$Q(\tau) = (q_{ground}, q_{path}, q_{verify}, q_{coh}, q_{eff})$
(decomposed quality vector, not a single score)

1.4 Necessity, and statistical analysis: Process Necessity $N(b)$ Measuring the Outcome Effects of Process Behaviors



1.5 Controlled experimental layer and task selection

 Main Audit experiments datasets: SWE-bench Pro Python subset (260 cases) and its small subset
Additionally datasets: SWE-bench Lite all and SWE-bench Verified part

 **Base model: GPT-5.6 Luna (max reasoning)**
Other model: GPT-4.1-mini, GPT-4.1, and GPT-5.1

RQ-driven experimental overview

RQ1: Quality vs. Necessity

 **Evidence Sources:**
(1.1&1.2) Successful traces
+
matched runs

Output:
 $Q(\tau)$ (process quality) × $N(b)$ (process necessity)

RQ2: Main Behavior Effects

 **Ablations:**
• Context • Search
Data:
• Components...
(1.5) SWE-bench Pro Python subset
Output:
 Δ localization / repair

RQ4: Robustness

 **Factors:**
• Representation • Policy
• Model • Replay
Output:
Stability across conditions

RQ3: Sufficiency under Cost

 **Data:**
(1.5) 40 tasks × 7 context arms
Metric:
Outcome + tokens / calls